MTH355

Timed Online Assignment – January Semester 2021

Basic Mathematical Optimization

Tuesday, 18 May 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Consider the following system of linear equations.

$$\begin{aligned} -2 + 3x_1 + 4x_2 - x_3 &= 0 \\ 19 + 12x_1 - 18x_2 + 6x_3 &= 0 \\ 5 - 3x_1 + 6x_2 + 9x_3 &= 0 \end{aligned}$$

(i) Write down the matrix A and vector b, where $Ax = b$. Apply the LU decomposition technique on A. Find the elementary matrices E_{ij}.

(6 marks)

(ii) Based on the LU of A in Question 1(a)(i) solve the system of linear equations.

(4 marks)

(iii) Use the LU of A to find the first column of A^{-1}.

(4 marks)

(iv) Use the LU of A to find the determinant of A.

(2 marks)

(b) Consider the system of linear equations.

$$\begin{aligned} 4x_1 + x_2 + 2x_3 &= -2 \\ 3x_1 + x_2 + 7x_3 &= 2 \\ x_1 + 3x_2 + x_3 &= 3 \end{aligned}$$

(i) Apply the conditions of convergence for a set of linear equations. Starting from $(x_1, x_2, x_3) = (a, 0,0)$, where a is a real number, perform two iterations of the Jacobi iterative scheme. Keep the values in terms of a. Provide enough details for the iterative process.

(5 marks)

(ii) Starting from $(x_1, x_2, x_3) = (1,1,1)$, perform three iterations of the Gauss-Seidel iterative scheme. Keep the values up to five decimal places. Provide enough details for the iterative process.

(4 marks)

Question 2

(a) Consider the following linear programming problem.

$$\begin{aligned} \text{Minimize } \ z &= -x_1 + x_2 \\ 3x_1 + x_2 &\leq 1 \\ 7x_1 + 3x_2 &\geq 4 \\ x_1 \text{ is free, } \ x_2 &\geq 0 \end{aligned}$$

(i) Analyse the linear programming problem using the two-phase simplex method. Find the optimal solution.

(13 marks)

(ii) Is this problem solvable using the dual simplex method? Provide an explanation.

(3 marks)

(b) Consider the following linear programming problem.

$$\begin{aligned} \text{Maximize } \ z &= 2x_1 + ax_2 \\ 4x_1 + 3x_2 &\leq 12 \\ 3x_1 + 4x_2 &\leq 12 \\ x_1, x_2 &\geq 0 \end{aligned}$$

where $a > 0$ is a real number.

(i) Using a graphical method or any other method, find a range of values for a such that the maximum value of z occurs at $(x_1, x_2) = \left(\frac{12}{7}, \frac{12}{7}\right)$.

(6 marks)

(ii) A value of a is chosen from the range found in Question 2(b)(i). A simplex method is used to find the maximum value of z. What is the minimum number of iterations needed? Provide an explanation.

(3 marks)

Question 3

Consider the following linear programming problem.

$$\begin{gathered}\text{Maximize } z = 3x_1 + 4x_2 + 6x_3 + 5x_4 \\ 7x_1 + 2x_2 + 6x_3 + 4x_4 \leq 7 \\ 8x_1 + 9x_2 + 5x_3 + 3x_4 \leq b_2 \\ x_1, x_2, x_3, x_4 \geq 0\end{gathered}$$

Let s_1 and s_2 denote the slack variables for the respective constraints. When $b_2 = 9$, the optimal tableau is given by Table Q3.

Basic	$\boldsymbol{x_1}$	$\boldsymbol{x_2}$	$\boldsymbol{x_3}$	$\boldsymbol{x_4}$	$\boldsymbol{s_1}$	$\boldsymbol{s_2}$	$\boldsymbol{v}$
$\boldsymbol{z}$	-63/10	0	-8/5	0	-11/10	-1/5	-19/2
$\boldsymbol{x_4}$	47/30	0	22/15	1	3/10	-1/15	3/2
$\boldsymbol{x_2}$	11/30	1	1/15	0	-1/10	2/15	1/2

Table Q3

(a) We allow b_2 to change from 9 to $(9 + a)$. Find a range of values for a in which x_4 and x_2 remain basic.

(4 marks)

(b) When b_2 changes from 9 to 34, modify the original optimal tableau. Solve the new linear programming problem.

(9 marks)

(c) We allow c_1, the coefficient of x_1 in the objective function, to change from 3 to $(3 + d)$. Find a range of values for d in which the optimal solution of the original problem remains unchanged.

(3 marks)

(d) When c_1, the coefficient of x_1 in the objective function, changes from 3 to 19/2, compute the new optimal solution.

(9 marks)

Question 4

(a) WheatCo, a wheat processing company, imports raw wheat from three countries X, Y, and Z to produce different grades (i.e., regular, first class, and premium) of wheat for its customers. After processing, the yields per barrel of wheat for different grades are shown in the last three columns of Table Q4(a). The second column shows the maximum supply (barrels/day) from each country. The third column shows the cost of raw wheat per barrel purchased from each country. WheatCo must supply a total of 4000, 3500, and 2100 barrels of regular, first class, and premium wheat, respectively, to its customers.

			Yield		
Country	**Supply (barrels/day)**	**Cost ($/barrel)**	**Regular**	**First class**	**Premium**
X	2500	150	0.2	0.3	0.4
Y	3000	180	0.3	0.2	0.3
Z	1800	210	0.3	0.4	0.1

Table Q4(a)

Formulate the wheat problem as a linear programming model with an aim to minimize the production cost while supply just enough wheat to its customers.

(7 marks)

(b) Consider the following integer linear programming model.

$$\text{Maximize } z = 3x_1 + 4x_2$$
$$35x_1 + 31x_2 \leq 217$$
$$x_1 + \frac{5}{3}x_2 \leq 8$$
$$x_1, x_2 \geq 0$$
$$x_1, x_2 \text{ are integers.}$$

(i) Determine graphically, the solution to the continuous problem.

(5 marks)

(ii) Employ the branch and bound method to solve the integer linear programming model. Draw the tree diagram for the solution process. Construct a table of results using the following format.

Problem, I_k	**Problem solved, C_k**	**Solution to C_k**	**Current bound**	**Problem to pursue**	**Stored problem**

<u>Table Q4(b)(ii)</u>

(13 marks)

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